확률·랜덤프로세스 기말대비　선별 12문항 — L05~L09 핵심 + 예측 보강 (★=기말 예측 시험지 직결)

① 오답노트.md 어제 오답부터 재풀이　② ★ 표시는 기말 예측 시험지 직결 문항(우선)　③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- 혼합형 RV: $F_X=C(x)+D(x)$, $E[X]=\int x\cdot c(x)dx + \Sigma\, x_k\cdot a_k$ (도약높이 a_k)
- 변환 $Y=g(X)$: CDF 먼저 → 미분; 단조구간 $f_Y=f_X(g\ ^{1}(y))\cdot|d\ g\ ^{1}/dy|$
- 결합: 주변=합/적분, 조건부=결합/주변, 독립　결합=주변곱
- 공분산 $Cov=E[XY]-E[X]E[Y]$, $\rho=Cov/(\sigma_X\ \sigma_Y)$, $Var(X+Y)=VarX+VarY+2Cov$
- 조건부 기댓값 $E[Y|X=x]=\int\ y\cdot f_{Y|X}(y|x)\ dy$
- CLT: ΣX　$N(n\mu, n\sigma^2)$; 표본평균 $E[M\ ]=\mu$, $Var(M\ )=\sigma^2/n$ (WLLN)
- 모든 PMF/PDF는 명시 구간 밖에서 otherwise = 0

문항 목차 (총 12문항)

한 문제당 1페이지 · 가로모드 · 풀이 공간 포함　|　권장(75분): ★우선 → 나머지 → 채점·오답등록

Problem 13

Let X be a random variable with the following CDF

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \le x < \frac{1}{4} \\ x + \frac{1}{2} & \text{for } \frac{1}{4} \le x < \frac{1}{2} \\ 1 & \text{for } x \ge \frac{1}{2} \end{cases}$$

a. Plot $F_X(x)$ and explain why X is a mixed random variable.
b. Find $P(X \le \frac{1}{3})$.
c. Find $P(X \ge \frac{1}{4})$.
d. Write CDF of X in the form of

$$F_X(x) = C(x) + D(x),$$

where $C(x)$ is a continuous function and $D(x)$ is in the form of a staircase function, i.e.,

$$D(x) = \sum_k a_k u(x - x_k).$$

e. Find $c(x) = \frac{d}{dx} C(x)$.
f. Find EX using $EX = \int_{-\infty}^{\infty} x c(x) dx + \sum_k x_k a_k$

— 풀이 —

Problem 1
Consider two random variables X and Y with joint PMF given in Table 5.4

Joint PMF of X and Y in Problem 1

	$Y = 1$	$Y = 2$
$X = 1$	$\frac{1}{3}$	$\frac{1}{12}$
$X = 2$	$\frac{1}{6}$	0
$X = 4$	$\frac{1}{12}$	$\frac{1}{3}$

a. Find $P(X \leq 2, Y > 1)$.
b. Find the marginal PMFs of X and Y.
c. Find $P(Y = 2 | X = 1)$.
d. Are X and Y independent?

— 풀이 —

Problem 18
Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x,y) = \begin{cases} \frac{1}{4}x^2 + \frac{1}{6}y & -1 \le x \le 1, 0 \le y \le 2 \\ \\ 0 & \text{otherwise} \end{cases}$$

a. Find the marginal PDFs, $f_X(x)$ and $f_Y(y)$.
b. Find $P(X > 0, Y < 1)$.
c. Find $P(X > 0 \text{ or } Y < 1)$.
d. Find $P(X > 0 | Y < 1)$.
e. Find $P(X + Y > 0)$.

— 풀이 —

5.5.1 Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/2 & -1 \le x \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch the region of nonzero probability and answer the following questions.

(a) What is $\mathrm{P}[X > 0]$?

(b) What is $f_X(x)$?

(c) What is $\mathrm{E}[X]$?

— 풀이 —

Problem 20
Let $X \sim N(0,1)$.

a. Find the conditional PDF and CDF of X given $X > 0$.
b. Find $E[X|X > 0]$.
c. Find Var$(X|X > 0)$.

— 풀이 —

6.2.1● The voltage X across a 1 Ω resistor is a uniform random variable with parameters 0 and 1. The instantaneous power is $Y = X^2$. Find the CDF $F_Y(y)$ and the PDF $f_Y(y)$ of Y.

— 풀이 —

Problem 31
Consider two random variables X and Y with joint PMF given in Table 5.6.

Table 5.6: Joint PMF of X and Y in Problem 31

	$Y=0$	$Y=1$	$Y=2$
$X=0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$
$X=1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$

Find $Cov(X,Y)$ and $\rho(X,Y)$.

— 풀이 —

5.7.5 ● X and Y are random variables with $\mathrm{E}[X] = \mathrm{E}[Y] = 0$ and $\mathrm{Var}[X] = 1$, $\mathrm{Var}[Y] = 4$ and correlation coefficient $\rho = 1/2$. Find $\mathrm{Var}[X + Y]$.

— 풀이 —

7.5.7 Over the circle $X^2 + Y^2 \le r^2$, random variables X and Y have the uniform PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/(\pi r^2) & x^2 + y^2 \le r^2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is $f_{Y|X}(y|x)$?

(b) What is $\mathrm{E}[Y|X = x]$?

— 풀이 —

Problem 5

The amount of time needed for a certain machine to process a job is a random variable with mean $EX_i = 10$ minutes and $\mathrm{Var}(X_i) = 2$ minutes^2. The times needed for different jobs are independent from each other. Find the probability that the machine processes less than or equal to 40 jobs in 7 hours.

— 풀이 —

10.1.2 ● $X_1, \ldots, X_n$ are independent uniform random variables with expected value $\mu_X = 7$ and variance $\mathrm{Var}[X] = 3$.

(a) What is the PDF of X_1?

(b) What is $\mathrm{Var}[M_{16}(X)]$, the variance of the sample mean based on 16 trials?

(c) What is $\mathrm{P}[X_1 > 9]$, the probability that one outcome exceeds 9?

(d) Would you expect $\mathrm{P}[M_{16}(X) > 9]$ to be bigger or smaller than $\mathrm{P}[X_1 > 9]$?

To check your intuition, use the central limit theorem to estimate $\mathrm{P}[M_{16}(X) > 9]$.

— 풀이 —

9.5.3● The duration of a cellular telephone call is an exponential random variable with expected value 150 seconds. A subscriber has a calling plan that includes 300 minutes per month at a cost of \$30.00 plus \$0.40 for each minute that the total calling time exceeds 300 minutes. In a certain month, the subscriber has 120 cellular calls.

(a) Use the central limit theorem to estimate the probability that the subscriber's bill is greater than \$36. (Assume that the durations of all phone calls are mutually independent and that the telephone company measures call duration exactly and charges accordingly, without rounding up fractional minutes.)

(b) Suppose the telephone company does charge a full minute for each fractional minute used. Re-calculate your estimate of the probability that the bill is greater than \$36.

— 풀이 —